ISR treatment in the hadronic W mass analysis

$e^+e^- \rightarrow W^+W^- \rightarrow qqqq$ **Centre of mass energies: 162.5, 240 and 365 GeV**

Supervisors: Kazem Azizi, Saeedeh Rostami, Matteo Defranchis, David D'Enterria

Group Members:
Mohadeseh Dadashzadeh, Shima Osloub, Amir Hossein Ebrahimnejad, Nazanin Zahra Norouzi

Method based on Marina Béguin, Calorimetry and W mass measurement for future experiments, Chapter 7

SCOPE

ISR treatment in the W mass fit

1. **Run the standard fit** — 4C conserves four momentum. 5C also imposes equal dijet masses.
2. **Identify a poor standard fit** — An ISR refit is attempted when the corresponding fit probability is below 3%.
3. **Add one collinear photon** — The photon represents beam energy lost through ISR or beamstrahlung.
4. **Check the physical behavior** — We study masses, probabilities, pulls and the relation to generator level ISR.

The analysis follows the reference method. The detector model and data samples are not identical to the reference study.

Why lost beam energy matters

Standard hypothesis

$$\sum_{\text{jets}} \vec{p} = 0, \qquad \sum_{\text{jets}} E = E_{\text{cm}}$$

If an ISR photon escapes along the beam pipe, the visible jets carry less than √s. The standard fit must then force the missing energy into the jets.

ISR aware hypothesis

$$\sum_{\text{jets}} \vec{p} + \vec{p}_\gamma = 0, \qquad \sum_{\text{jets}} E + E_\gamma = E_{\text{cm}}$$

The photon provides a physical way to account for missing longitudinal energy while the jet parameters remain constrained by their measured covariance.

For 5C, the equality of the two fitted dijet masses is imposed in addition to four momentum conservation.

One effective photon along the beam axis

$$p_{T,\gamma} = 0, \qquad E_\gamma = |p_{z,\gamma}|$$

The model adds one degree of freedom. The photon may travel along either beam direction, but it has no transverse momentum.

Fit	Photon	Equal mass
4C	No	No
4C + ISR	Yes	No
5C + ISR	Yes	Yes

4C + ISR tests the photon model without the equal mass condition. 5C + ISR solves the same problem with one additional constraint.

From the fitted variable y to photon momentum

$$b = \frac{2\alpha_{\mathrm{em}}}{\pi}\left[\ln\left(\frac{s}{m_e^2}\right) - 1\right]$$

$$p_{z,\gamma}(y) = \mathrm{sign}(y)\, E_{\max}\left[\mathrm{erf}\left(\frac{|y|}{\sqrt{2}}\right)\right]^{1/b}$$

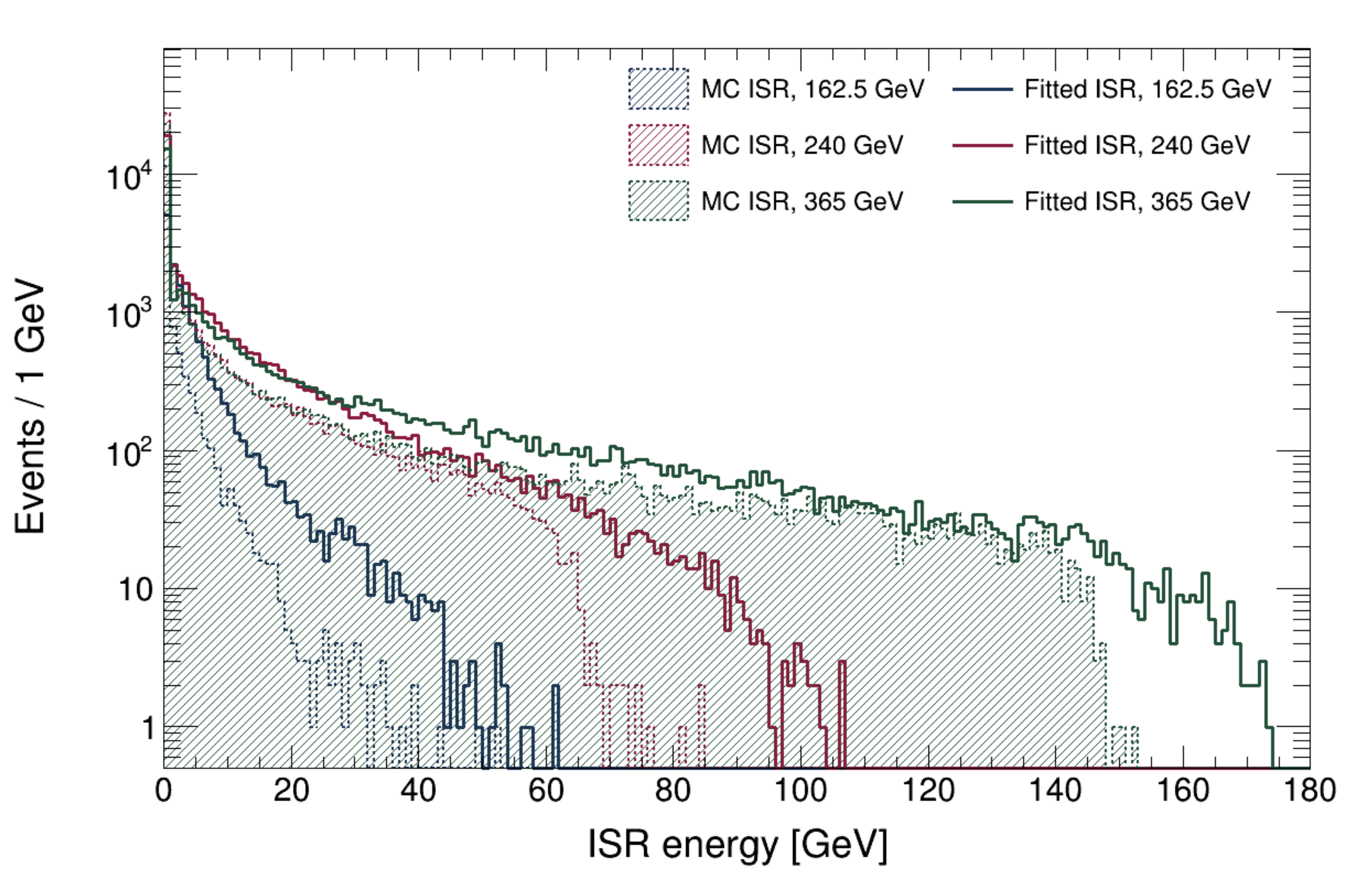
$$E_{\max} = \frac{\sqrt{s}}{2}\left(1 - \frac{M_W^2}{s}\right)$$

Here y ~ N(0,1) is the fitted coordinate. Second equation maps y to the signed longitudinal photon momentum; p_{T,γ} = 0 and E_γ = |p_{z,γ}|.

Emax?

Energy	Single-W [GeV]	WW-threshold [GeV]
162.5	61.368	1.721
240	106.538	66.152
365	173.648	147.093

Fitted ISR follows the tail but extends beyond truth



What the figure shows

The fitted spectrum reaches the physical high energy tail at all three collision energies.
The fitted distribution is broader than generator level ISR.
A truth correlation test is needed before interpreting the fitted photon as ISR event by event.

The photon is allowed, but it is not free

$$\chi^2 = \Delta_{\text{jet}}^{\mathsf{T}} V_{\text{jet}}^{-1} \Delta_{\text{jet}} + y^2$$

Jet term

Measures how far the fitted jet parameters move from their reconstructed values, using the measured covariance matrix.

Photon term

The y^2 contribution is the Gaussian constraint. Large photon solutions remain possible, but they increase χ^2.

Without the y^2 term, the additional photon could absorb longitudinal imbalance with almost no cost.

When the ISR refit enters the analysis

Standard fit	Run 4C and 5C independently.
Probability test	Attempt the matching ISR refit if the standard fit converges and $P(\chi^2) < 3\%$.
ISR solution	Require convergence, finite masses and χ^2, and a constraint residual below 10^{-5}.
Final variables	Use the ISR result only when that refit is valid. Otherwise keep the standard result.

No event is removed at this stage. The producer stores standard and ISR results so that later selections can be studied explicitly.

The same ISR fit is started from two initial values

Start A

$$y = 0$$

No photon in the initial hypothesis

same physical model

Start B

$$y \text{ from raw missing p\_z}$$

Initial photon estimate from the measured longitudinal imbalance

Among the valid solutions, keep the one with the lower χ^2.

Validity checks and pull definition

Check	Requirement
Minimiser	Migrad status = 0
Constraints	max $\lvert f_k \rvert < 10^{-5}$
Output	finite χ^2 and masses

$$\mathrm{pull}_i = \frac{x_i^{\mathrm{fit}} - x_i^{\mathrm{meas}}}{\sigma_i^{\mathrm{change}}}$$

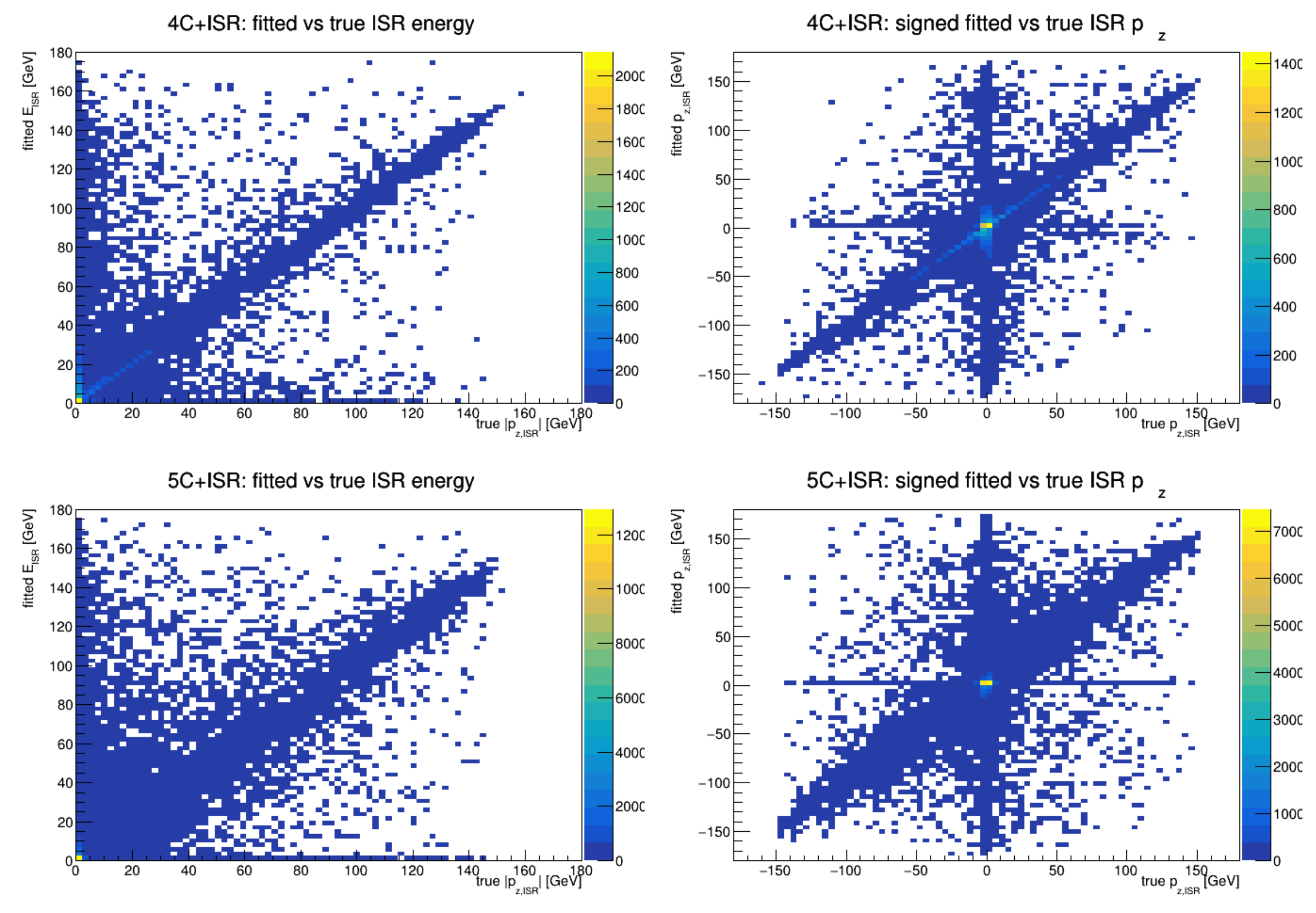
$$\left(\sigma_i^{\mathrm{change}}\right)^2 = (V_{\mathrm{meas}} - V_{\mathrm{fit}})_{ii}$$

$$V_{\mathrm{meas}} - V_{\mathrm{fit}} = VA^{\mathsf{T}}\left(AVA^{\mathsf{T}}\right)^{-1}AV$$

A is the constraint Jacobian evaluated at the fitted solution.

When ISR is used, the measured vector contains 16 jet parameters plus y, with V = diag(Vjet, 1). The reported pull is calculated from the fit actually used for that event.

ISR truth correlation is moderate overall



Reading all four panels

4C + ISR:
ρ(E) = 0.580
ρ(signed pz) = 0.679

5C + ISR:
ρ(E) = 0.595
ρ(signed pz) = 0.590
The diagonal tail is visible, while the low ISR region remains broad. The fitted photon is useful statistically, but not yet a precise event by event ISR estimator.

At 365 GeV, 5C ISR recovery remains low

4C + ISR

True ISR	Recovery	ρ(signed pz)
< 5 GeV	40.8%	0.011
5–20 GeV	68.6%	0.298
20–50 GeV	68.1%	0.678
≥ 50 GeV	62.9%	0.937

5C + ISR

True ISR	Recovery	ρ(signed pz)
< 5 GeV	15.3%	0.050
5–20 GeV	21.1%	0.323
20–50 GeV	19.2%	0.574
≥ 50 GeV	11.6%	0.811

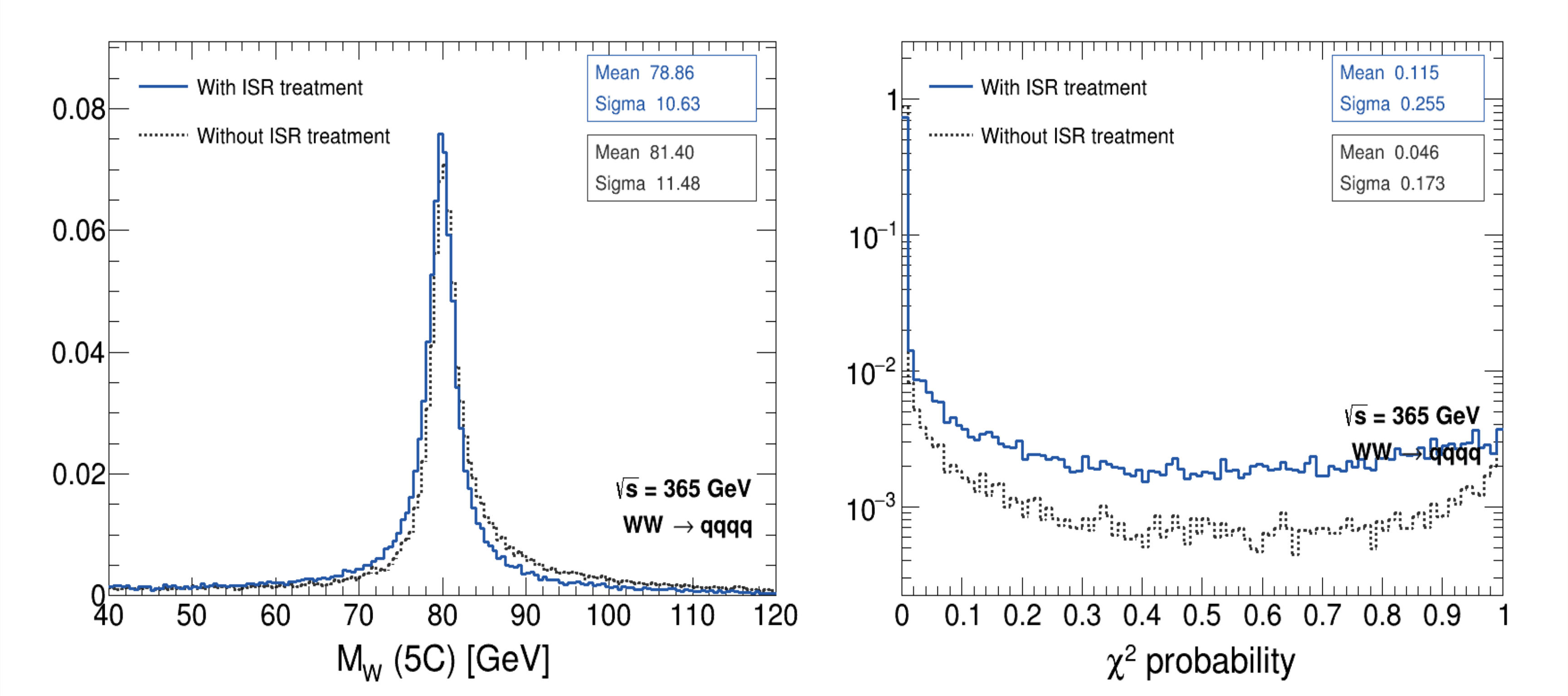
Recovery = final ISR probability > 3% / applied.
Overall: 4C 11,145 / 20,696 = 53.85%; 5C 6,548 / 40,744 = 16.07%.
Coverage (applied / standard-valid): 45.51% for 4C and 90.13% for 5C.

Open to discussion and current problems

Event topology	Is truth-level n_partons = 4 appropriate for selecting the reconstructed hadronic channel?
ISR model	What should ISR recovery provide near threshold, and should Emax use $1 - M\_W^2/s$ or $1 - 4M\_W^2/s$?
4C and 5C pulls	Why do α and x remain shifted before and after ISR treatment and are the 5C pull plots needed? Plot for every jet or all the jets together like the reference?
5C efficiency	Why does the final 5C selection keep only 24.1% of events at 365 GeV?

5C mass width: The final 5C σ increase from 3.44 to 17.01 GeV as the pre-5C mass difference grows.

ISR refitting improves the 5C probability



Mean mass

Current: 81.40 → 78.86 GeV
Reference: 82.18 → 79.00 GeV

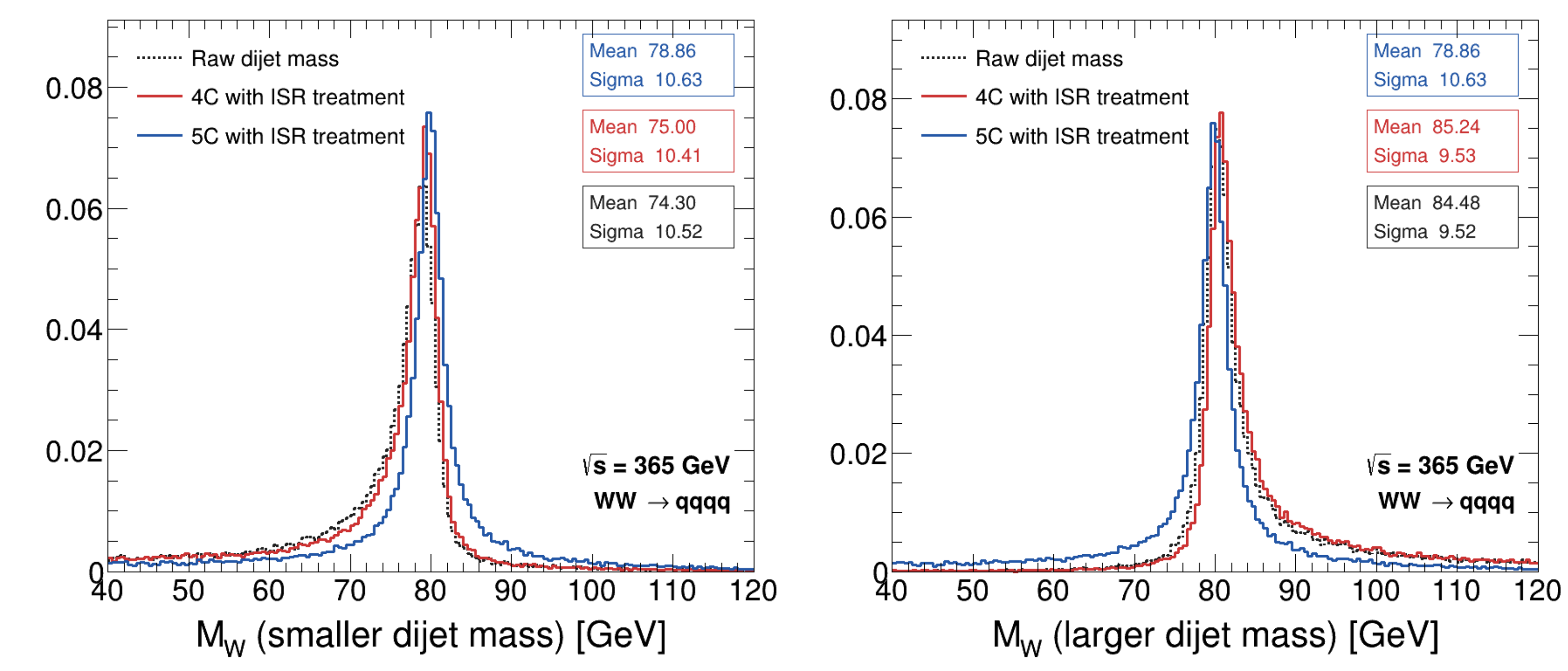
Mass width

Current: 11.48 → 10.63 GeV
Reference: 10.15 → 8.49 GeV

Mean probability

Current: 0.046 → 0.115
Reference: 0.0769 → 0.1249

4C and 5C constrain dijet masses differently



4C + ISR

Current: small 75.00 ± 10.41; large 85.24 ± 9.53 GeV

Reference: 74.62 ± 10.21; 85.24 ± 9.19 GeV

5C + ISR

Current: 78.86 ± 10.63 GeV

Reference: 79.00 ± 8.49 GeV

σ is 2.14 GeV broader than the reference.

Acceptance improves, but 5C ISR recovery remains low

4C acceptance and ISR recovery

Quantity	Standard P>3%	Final / recovery
162.5 GeV acceptance	69.5%	85.4%
240 GeV acceptance	63.4%	85.4%
365 GeV acceptance	54.5%	78.8%
365 GeV ISR recovery	11,145 / 20,696	53.85%

5C acceptance and ISR recovery

Quantity	Standard P>3%	Final / recovery
240 GeV acceptance	8.0%	22.3%
365 GeV acceptance	9.8%	24.1%
365 GeV ISR applied	40,744 events	—
365 GeV ISR recovered	6,548 / 40,744	16.07%

Final acceptance uses the hybrid standard/ISR result. ISR recovery efficiency = recovered / applied; recovered means final ISR probability > 3%.

The 5C width grows with the pre-5C mass difference

240 GeV

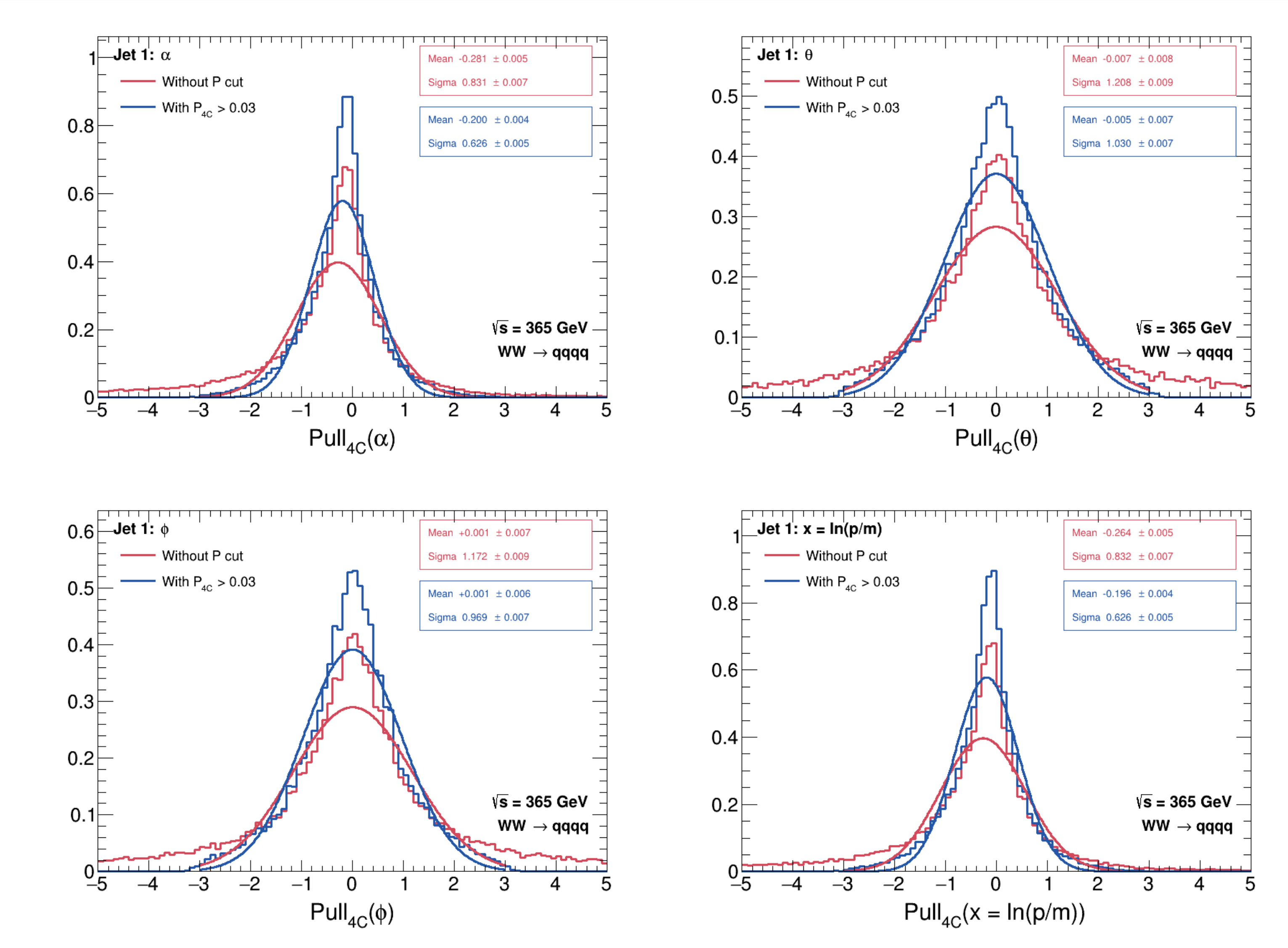
4C mass split	Final 5C fails	Final σ
0–2 GeV	42.2%	4.08 GeV
2–5 GeV	69.2%	5.16 GeV
10–20 GeV	92.4%	10.26 GeV
≥20 GeV	95.5%	14.59 GeV

365 GeV

4C mass split	Final 5C fails	Final σ
0–2 GeV	39.3%	3.44 GeV
2–5 GeV	69.8%	4.45 GeV
10–20 GeV	90.4%	10.03 GeV
≥20 GeV	93.4%	17.01 GeV

Both the rejection rate and the final 5C width rise sharply with Δm after 4C; ISR recovery does not remove this trend.

Jet 1 · standard 4C pulls without ISR at 365 GeV



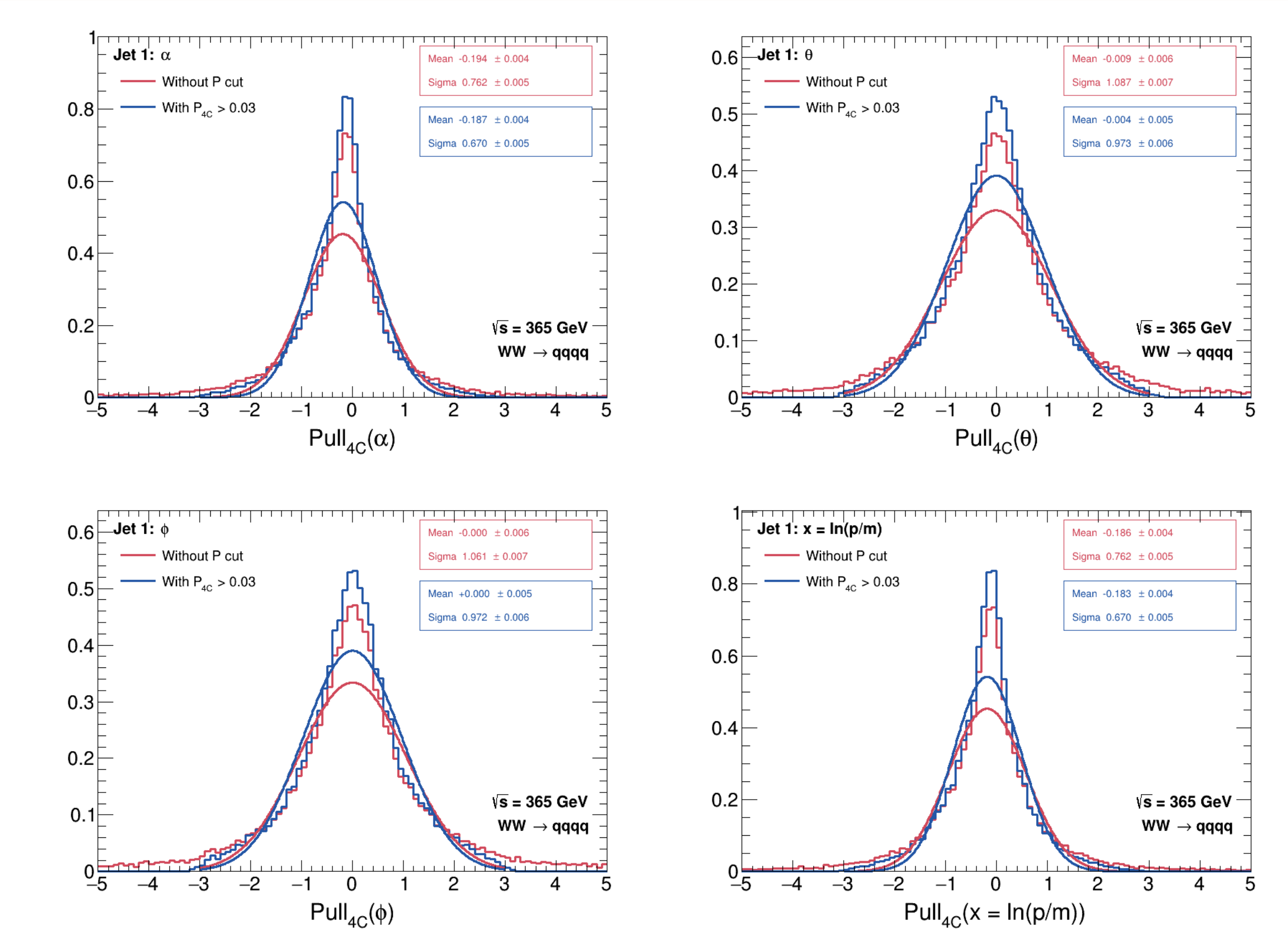
Standard 4C · jet 1

Red: all valid standard 4C fits.

Blue: standard 4C with P_4C > 0.03.

The probability cut narrows θ and φ and reduces the α and x tails, although α and x remain negatively shifted.

Jet 1 · final 4C pulls with ISR treatment at 365 GeV



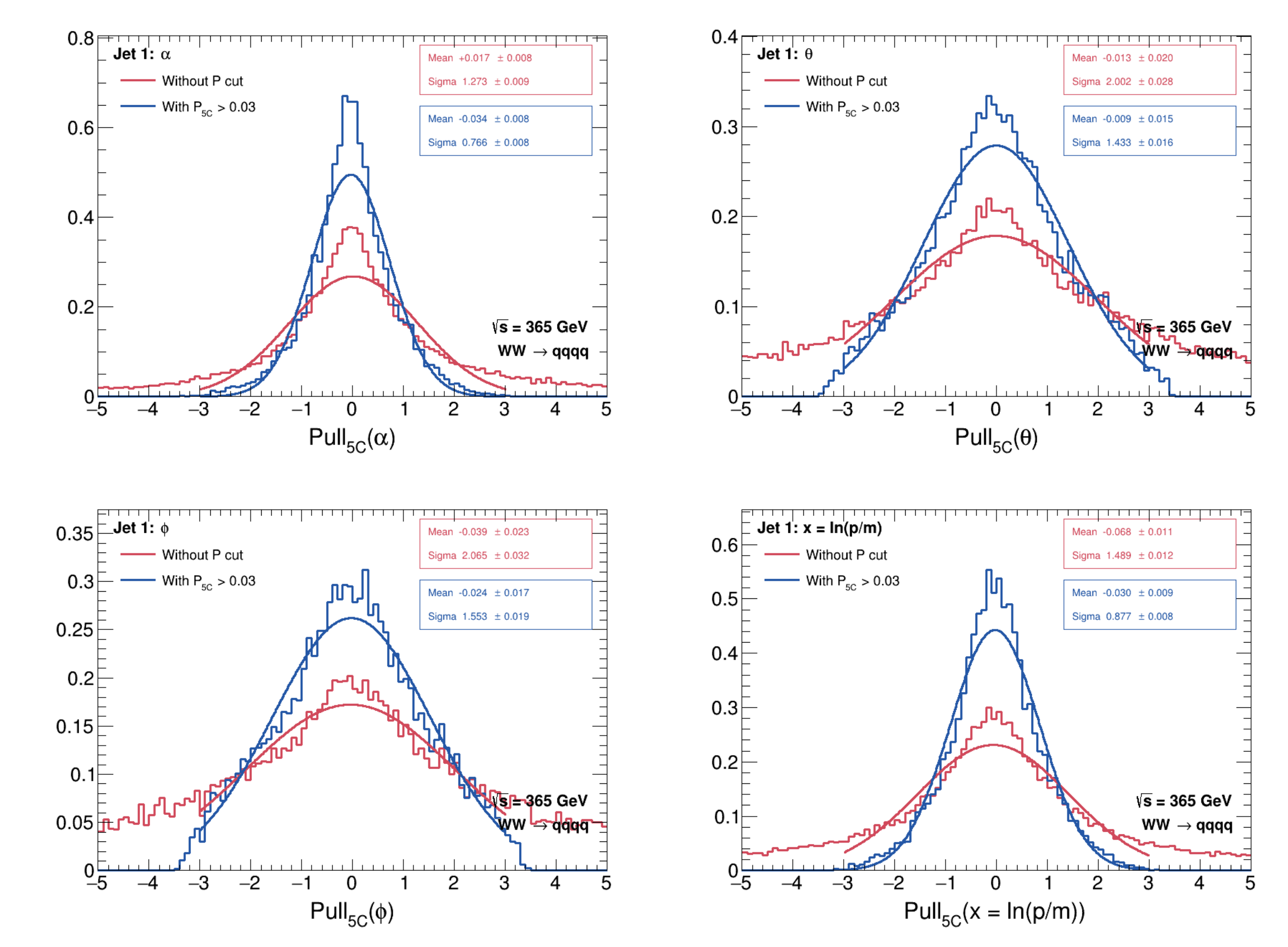
Final 4C · jet 1

Final means: keep the standard fit unless a low probability event has a valid 4C+ISR refit.

Blue applies P_final > 0.03.

σ_θ = 0.973 and σ_φ = 0.972. The α and x means remain near −0.18.

Jet 1 · final 5C pulls with ISR treatment at 365 GeV



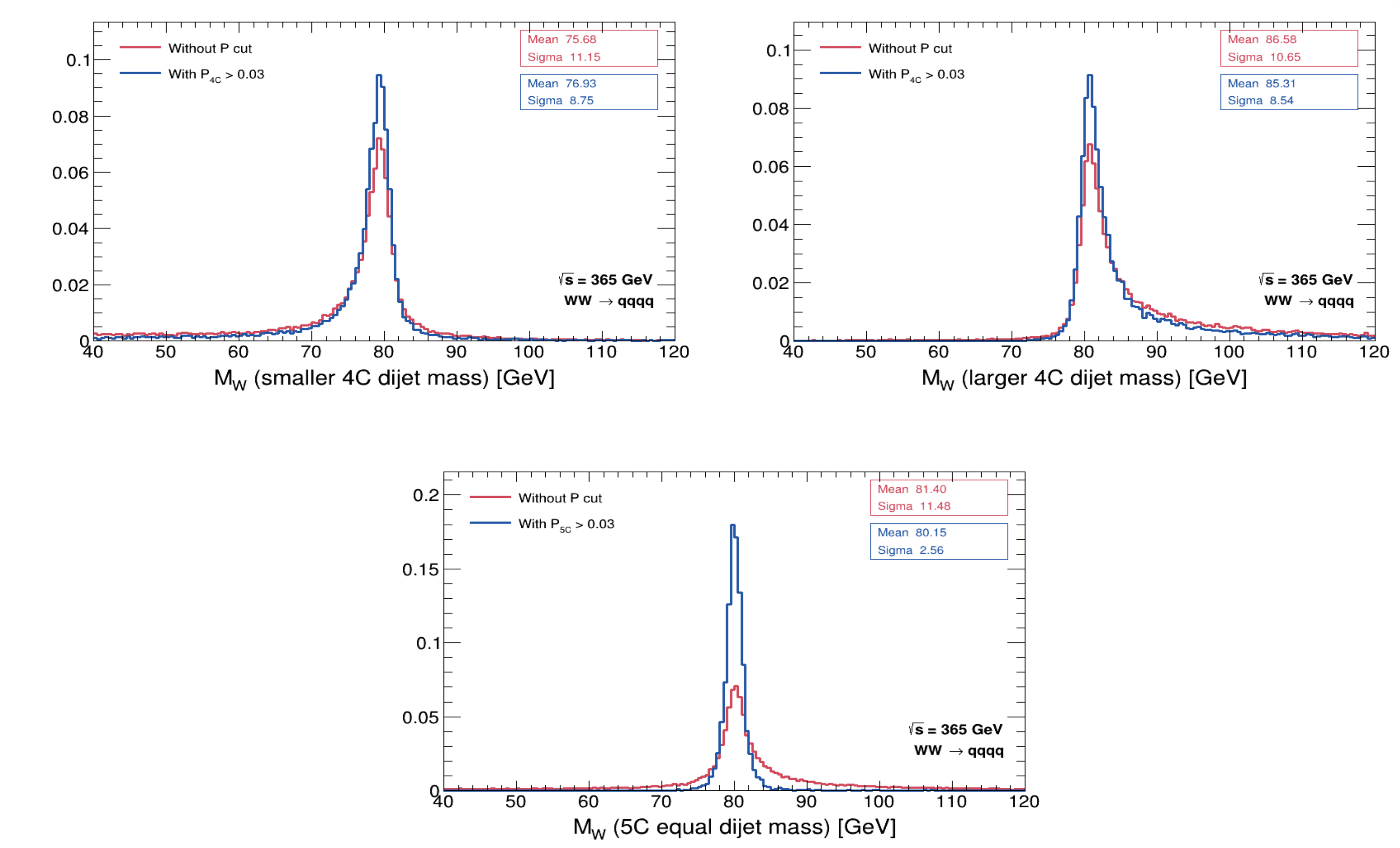
Final 5C · jet 1

Final means: keep the standard 5C fit unless a low probability event has a valid 5C+ISR refit.

Blue applies P_final > 0.03.

The angular pulls are broad: σ_θ = 1.433 and σ_φ = 1.553. This is an open issue.

P > 3% narrows the standard-fit W-mass distributions



365 GeV

Without ISR treatment

Current σ, no cut → P>3%:
4C small 11.15 → 8.75
4C large 10.65 → 8.54
5C 11.48 → 2.56 GeV

Reference Appendix B.4:
4C small 10.26 → 8.61
4C large 9.79 → 8.44
5C 10.15 → 2.70 GeV